

KOGANTI SASANK

Senior Data Scientist | Generative AI, Agentic Systems & Applied ML

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PROFESSIONAL SUMMARY

Senior Data Scientist with 4 years of experience designing and delivering production-grade Generative AI, agentic RAG and machine-learning solutions for procurement, industrial R&D and energy operations. Proven experience leading solution architecture, secure self-hosted LLM deployments, evaluation and observability, and scalable cloud-native systems. Combines hands-on engineering with project planning, mentoring and stakeholder leadership.

CORE SKILLS

GenAI & Agents: LangGraph, multi-agent orchestration, agentic RAG, QLoRA, few-shot prompting, LLM-as-judge, NeMo Guardrails

Engineering: Python, SQL, FastAPI, Flask, Docker, Kubernetes, KEDA, RabbitMQ, Celery, Redis

Data & ML: Pandas, NumPy, scikit-learn, XGBoost, PostgreSQL, MariaDB, Elasticsearch

Models & Retrieval: Gemma, Llama 3.1, Gemini, Ollama, vLLM, BM25, hybrid retrieval, Elasticsearch vector DB

MLOps & Observability: Jenkins CI/CD, MLflow, Kubernetes, Langfuse, pytest, Bandit, model quality gates

Leadership: Solution architecture, project planning, mentoring, GenAI hiring, guild leadership, stakeholder communication

PROFESSIONAL EXPERIENCE

Senior Data Scientist

Jul 2025 - Present

TCG Digital | Remote, India

- Architected an enterprise procurement intelligence platform with a supervisor and five specialized agents spanning RFQ tracking, bid-stage validation, post-PO fulfilment, compliance and user queries, unified SAP Ariba, and email workflows in a role-based dashboard interface.
- Designed a production agentic RAG architecture with SQL-first routing for deterministic queries, hybrid retrieval for document-grounded questions, small-model intent classification, LLM-based response evaluation and NeMo Guardrails for relevance, safety and information-leakage checks.
- Scaled asynchronous processing for up to 2,000 concurrent users using RabbitMQ, Celery, Redis, Kubernetes and KEDA, implemented Langfuse tracing and confidence scoring using retrieval similarity and answer-quality evaluation.
- Led CataLLM, a catalyst-research engine that screened 50K+ patents using BM25 and LLM scoring, curated high-value documents and fine-tuned Gemma with QLoRA to generate evidence-backed hypotheses, achieved 65% relevance and deployed on GCP.
- Built a private technical-proposal generation system using self-hosted Llama 3.1 served through vLLM and section-specific few-shot examples, reducing dependence on manual drafting while meeting data-security requirements.
- Lead the Agentic Innovation Guild, co-lead DevOps and Optimization guilds, mentored 5-6 colleagues, led two team members and interviewed 10-12 GenAI engineering candidates.

Data Scientist Consultant

Jul 2022 - Jun 2025

TCG Digital | Remote, India

- Created a reusable CI/CD framework for data-science APIs with automated linting, Bandit security scans, unit tests, golden-dataset model checks and Jenkins-based FastAPI/Flask deployment.
- Delivered XGBoost models for drilling-fluid rheology and thickening-time prediction across 1M+ records and 50+ locations, comparing global, local and grouped-location approaches, completed delivery in under two months.
- Built a reusable AutoML framework for feature engineering, model selection, validation and reporting, with optional TPOT, PyCaret and AutoGluon integration for an industrial analytics platform.
- Developed optimization and ML solutions for refinery and manufacturing operations, building experience in production integration, constrained decision systems and cross-functional delivery.

INDEPENDENT PROJECT

Agentic Recruitment Platform | LangGraph, Python, Gemma 2B, Ollama, React

- Architected and developed the backend for a deployed end-to-end recruitment product serving candidates, HR teams, hiring managers and business-unit leaders, with six agents for JD creation, resume screening, interview-question generation, traceability, audit and workflow automation.
- Self-hosted a quantized Gemma 2B model through Ollama for private inference and built the agent workflows in LangGraph, currently improving latency, resilience and production readiness. React UI was developed using AI-assisted coding.

Repository: github.com/kogantisasank831-ux/rms-app-optimized

EDUCATION & RECOGNITION

M.Sc. Mathematics, BITS Pilani | 2018 - 2022

Recognition: Spot Award and Team Spot Award for real-time optimization delivery.